

LINGLING, China

WMO: 578660

Lat: 26.2259N

Lon: 111.6165E

Elev: 571

StdP: 14.40

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 30.8	(c) 32.9	(d) 18.4	(e) 14.1	(f) 44.0	(g) 23.2	(h) 17.9	(i) 43.1	(j) 17.9	(k) 46.8	(l) 15.9	(m) 44.2	(n) 6.1	(o) 20	(p) 0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	96.3	77.8	94.3	77.8	92.4	77.5	81.2	88.0	80.2	87.9	79.4	87.5	6.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.0	159.3	83.1	78.5	151.2	83.0	77.3	145.3	83.0	45.7	87.4	44.5	88.1	43.6	87.9	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.5	(b) 14.1	(c) 12.2	(e) 29.2	(f) 99.4	(g) 2.2	(h) 2.2	(i) 27.6	(j) 101.0	(k) 26.4	(l) 102.2	(m) 25.1	(n) 103.4	(o) 23.5	(p) 105.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			29.2	99.4	2.2	2.2	27.6	101.0	26.4	102.2	25.1	103.4	23.5	105.0		
			27.3	82.2	2.2	1.8	25.7	83.5	24.5	84.6	23.2	85.6	21.6	87.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.6	43.9	48.1	55.5	66.3	73.9	79.8	84.9	83.3	77.0	68.0	57.9	47.9
	DBStd	15.54	7.37	9.25	8.74	8.16	5.90	4.92	3.99	4.75	6.34	6.36	7.47	6.05
	HDD50	527	219	138	40	2	0	0	0	0	0	0	16	112
	HDD65	2352	655	480	317	82	5	0	0	0	3	42	236	532
	CDD50	6228	30	84	210	489	742	895	1081	1033	810	558	251	45
	CDD65	2579	1	6	23	120	281	445	616	568	363	135	21	0
	CDH74	25881	1	35	128	833	2165	4377	7699	6546	3248	732	115	2
	CDH80	10745	0	4	15	210	702	1672	3734	2974	1244	170	20	0
(14) Wind	WSAvg	5.8	5.9	6.2	5.9	6.0	5.5	5.5	5.9	5.6	5.8	5.6	5.7	5.9
(15) Precipitation	PrecAvg	53.2	2.6	3.6	5.0	8.2	8.6	6.6	3.4	4.8	2.6	3.0	3.1	1.9
	PrecMax	76.3	5.3	8.2	10.0	17.4	18.6	11.0	9.6	12.0	7.2	6.5	7.7	3.9
	PrecMin	37.4	0.2	0.8	1.7	3.7	3.0	1.8	0.2	0.1	0.0	0.0	0.1	0.0
	PrecStd	9.9	1.5	1.8	2.3	3.3	4.0	2.5	2.3	2.8	2.1	1.9	2.2	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.6	78.4	81.9	89.4	92.3	94.6	98.8	99.2	96.2	89.4	83.2	70.3
		MCWB	60.1	64.2	69.1	71.7	75.2	78.1	77.6	77.8	77.2	73.3	68.0	57.5
	2%	DB	63.8	73.4	77.6	85.6	89.5	92.0	96.3	96.6	93.1	84.6	77.2	64.7
		MCWB	53.9	61.6	65.9	71.8	74.3	78.0	78.1	77.8	76.9	70.4	64.8	54.0
	5%	DB	59.2	68.4	73.8	82.1	86.8	90.0	94.4	94.2	90.2	81.2	73.2	60.7
		MCWB	51.0	59.2	64.1	70.2	73.8	77.6	77.9	77.8	75.9	68.4	62.6	51.0
	10%	DB	54.6	62.2	69.3	79.0	84.0	87.9	92.5	91.8	87.1	78.1	69.2	57.3
		MCWB	48.2	54.6	61.0	69.0	73.5	77.3	77.9	77.7	74.8	67.0	59.7	49.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.4	65.9	71.0	75.4	80.3	82.0	83.3	82.1	80.0	76.1	70.4	61.7
		MCDB	68.6	76.3	79.6	83.8	86.3	86.4	88.3	88.0	89.3	85.6	78.8	66.4
	2%	WB	56.3	63.0	68.3	73.6	78.0	80.3	81.6	80.8	78.8	73.3	67.3	56.8
		MCDB	61.7	71.9	75.5	81.9	84.9	86.7	88.2	88.4	87.4	79.8	74.8	62.6
	5%	WB	52.0	59.9	65.5	72.2	76.4	79.2	80.4	79.9	77.7	71.3	64.8	53.3
		MCDB	57.5	67.7	72.0	80.0	83.4	86.2	88.4	88.1	86.3	77.5	70.7	57.9
	10%	WB	48.8	55.5	62.3	70.5	74.9	78.3	79.4	79.1	76.5	69.4	61.8	50.8
		MCDB	54.0	61.2	68.3	77.4	81.5	85.3	88.0	87.7	84.7	75.5	66.8	55.3
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.5	10.3	11.7	12.0	10.9	12.2	12.4	12.1	11.7	11.3	10.2
		MCDBR	17.9	20.2	18.7	17.9	17.3	13.8	15.5	16.4	17.0	17.3	18.0	18.9
	5% WB	MCWBR	9.9	10.9	9.6	7.4	6.7	4.8	4.1	4.0	4.9	6.1	7.8	9.6
		MCDBR	15.3	18.7	16.4	15.1	13.2	11.8	13.7	13.7	14.5	14.0	14.6	14.0
		MCWBR	9.1	11.0	9.9	7.9	6.7	5.3	5.2	4.6	4.8	6.5	7.6	8.3
(40) Clear-Sky Solar Irradiance	taub		0.691	0.719	0.790	0.753	0.653	0.623	0.563	0.651	0.692	0.745	0.668	0.665
	taud		1.502	1.494	1.416	1.520	1.741	1.823	1.989	1.743	1.635	1.492	1.593	1.563
	Ebn at Noon		177	187	185	199	220	225	239	218	203	179	181	176
	Edh at Noon		82	89	100	92	74	68	57	73	79	87	74	74
(44) All-Sky Solar Radiation	RadAvg		531	630	778	1006	1159	1246	1632	1424	1189	971	782	644
	RadStd		131	169	117	141	176	189	183	110	121	134	136	157

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.70	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	+261	+149
(47) Regional (7 neighbors)		+0.49	N/A	N/A	+0.86	N/A	N/A	N/A	N/A	+214	+128

Nomenclature: See separate page